

TrES-5b

R-1363

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_141020 · created 2026-07-31T23:53:43+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456950.6986 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.031
Transit depth (Rp/Rs)^2	2.71 +/- 1.04 [%]
Orbital Inclination (inc)	83.88 +/- 2.11
Airmass coefficient 1 (a1)	1.014 +/- 0.026
Airmass coefficient 2 (a2)	-0.009 +/- 0.023
Scatter in the residuals of the lightcurve fit is	2.55 %
Best Comparison Star	#1 - [193, 354]
Optimal Method	PSF photometry
Transit Duration (day)	0.071 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.165 ± 0.031 vs 0.142 (deviation 0.74σ)
Duration fit vs published	0.071 d vs 0.0758 d (deviation 0.35σ)
Mid-time O–C	0.7 min
Depth SNR	5.3
Residual scatter	2.55 %

Light curve

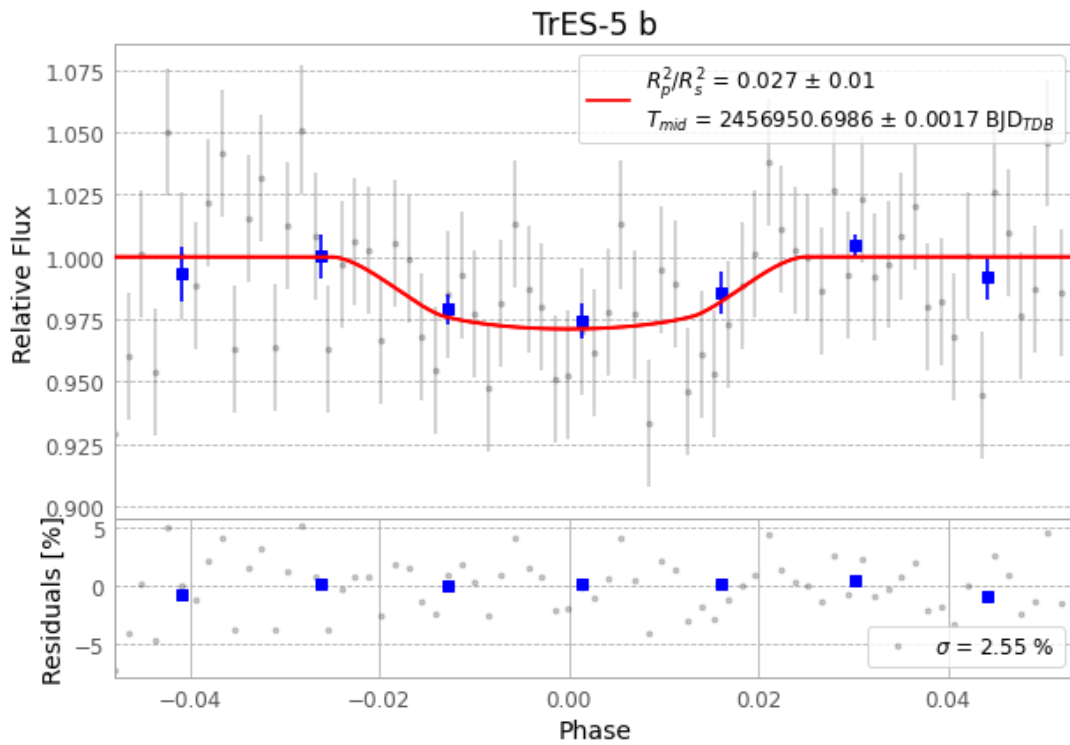


Figure 1 — FinalLightCurve_TrES-5 b_2014-10-19.png (SHA-256 39c686a7f91a101f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 321], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b0985218d6b15359...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), TRES-5141020024812.FITS → TRES-5141020063612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-141020.log (4db3794e02d7...), FinalLightCurve_TrES-5 b_2014-10-19.png (39c686a7f91a...), FinalLightCurve_TrES-5 b_2014-10-19.csv (0f3475b4d95e...)

Compute resources

Wall time 255.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 23:53Z.